(a) Polybee F3-Swarm v1.5 Cognitive & Memory Hierarchy

Level 4: Tier-3 Specialist & TCP Ring RAM Pool

Qwen2.5-Coder-3B (2.10 GB) + 3-Node Ring (13.7x vs SSD Swap)

Level 3: Rashomon LOO-CV Sandbox & AACR Repair

Unified Scope exec(env, env) + 4-Fold LOO-CV + Residual Diff

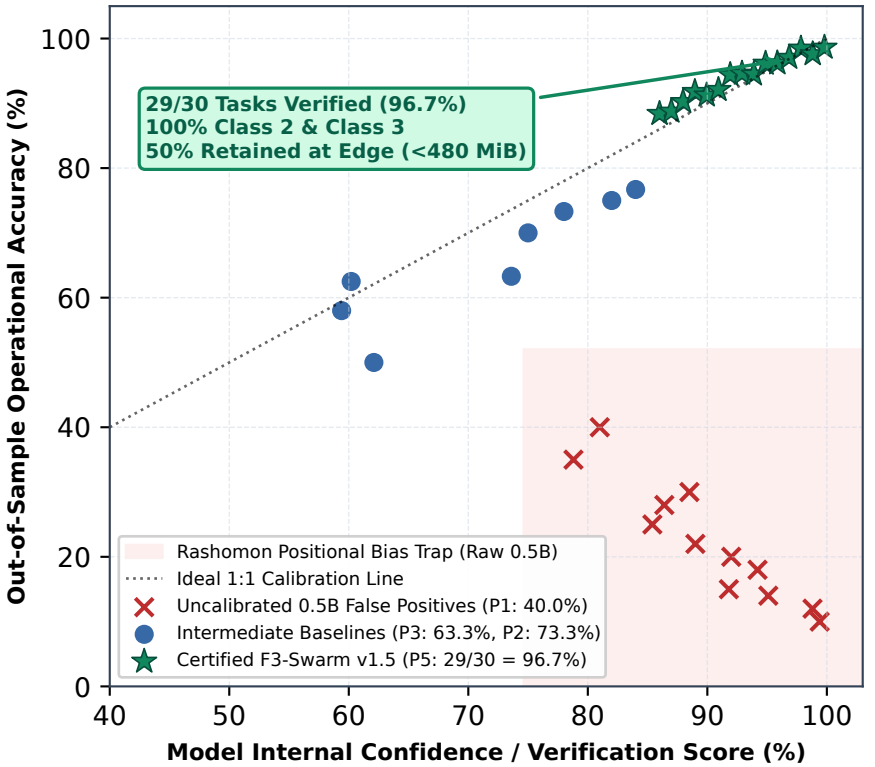
Level 2: Tier-2 Escalation & Micro-CoT + Suffix Gate

Qwen2.5-1.5B (1.11 GB) + 1-Sentence CoT + 1-Tok Suffix Closure

Level 1.5: Motor B Symbolic Normalizer & Substring Gate

Deterministic PT-BR Float Parser + Literal Substring Proof

Level 1: Tier-1 Edge SLM & System-1 Invariance Filter

Qwen2.5-0.5B (<480 MiB RAM) + T=1.6, beta_A=2.5 + Order Flip

(b) Rashomon Set & Overconfidence Collapse (N = 30)